

A Smooth Counterexample to the Exponential Total-Variation Conjecture for the Truncated Hilbert Transform

Automatic mathematics run `fa_banach_001`

9 August 2026

Status. Candidate counterexample, likely valid, for expert review. The construction gives a full negative answer to equation (2.1) and to the exponential stability conjecture after Theorem 2.1 of Alaifari–Pierce–Steinerberger. It works for every interval pair in their Cases 3 and 4 and already uses real functions in $C_c^\infty(I)$.

1 Source conjecture

For finite open intervals $I, J \subset \mathbb{R}$, the truncated Hilbert transform is normalized here as

$$Hf(x) = \frac{1}{\pi} \text{p.v.} \int_I \frac{f(y)}{x-y} dy, \quad x \in J.$$

The source is R. Alaifari, L. B. Pierce, and S. Steinerberger, *Lower bounds for the truncated Hilbert transform*, arXiv:1311.6845, published in *Revista Matemática Iberoamericana* 32 (2016), 23–56 [1].

On printed page 7, Section 2.2, for I, J in Case 3 (a gap) or Case 4 (overlap), Theorem 2.1 establishes the existence of some positive stability function for $BV(I)$ and then conjectures an exponential lower rate. Equation (2.1) states the corresponding concrete inequality for weakly differentiable functions:

$$\|Hf\|_{L^2(J)} \geq c_1 \exp\left(-c_2 \frac{\|f'\|_{L^1(I)}}{\|f\|_{L^2(I)}}\right) \|f\|_{L^2(I)}, \quad (1)$$

where $c_1, c_2 > 0$ may depend on I, J but not on f . The source evidence is reproduced below.

7

and thus identify the total variation with $\|f_x\|_{L^1(I)}$. In light of Theorem 2.1 the total variation seems to be the natural quantity with which to track the behavior of regular functions, and we conjecture that

$$(2.1) \quad \|Hf\|_{L^2(J)} \geq c_1 \exp\left(-c_2 \frac{\|f_x\|_{L^1(I)}}{\|f\|_{L^2(I)}}\right) \|f\|_{L^2(I)}.$$

An inequality of this form would quantify the physically intuitive notion that tomographic reconstruction is more difficult for inhomogeneous objects with high variation in density than it is for relatively uniform objects. Our first result toward this conjecture considers $\|f_x\|_{L^2(I)}$ instead, which provides access to Hilbert space techniques that allow us to prove the following statement:

The paper itself notes in Sections 1.5 and 2.4 that Case 4 contains the gap mechanism by restricting to functions compactly supported away from J . That is precisely the localization used below.

2 Proof intuition and result

For $K > 0$, define the maximal lower-stability envelope

$$\mathcal{S}_{I,J}(K) = \inf \left\{ \frac{\|Hf\|_{L^2(J)}}{\|f\|_{L^2(I)}} : 0 \neq f \in W^{1,1}(I), \quad \frac{\|f'\|_{L^1(I)}}{\|f\|_{L^2(I)}} \leq K \right\}. \quad (2)$$

The conjecture would imply $\mathcal{S}_{I,J}(K) \geq c_1 e^{-c_2 K}$.

The counterexample is a train of disjoint smooth bumps with coefficients

$$1, -\binom{m}{1}, \binom{m}{2}, \dots, (-1)^m.$$

The L^2 norm of this train is governed by $\sum_j \binom{m}{j}^2 = \binom{2m}{m}$, whereas its total variation is governed by $\sum_j \binom{m}{j} = 2^m$. Their quotient contributes only $m^{1/4}$. Each bump has width comparable to m^{-1} , contributing another $m^{1/2}$, so the normalized variation is $O(m^{3/4})$.

On the observation interval the Hilbert kernel is smooth. The alternating binomial coefficients are an m th finite difference, hence annihilate the first $m-1$ orders of the kernel. The remainder is geometrically small in m . Thus the data are $O(e^{-cm})$ while the regularity parameter is only $O(m^{3/4})$, producing an $e^{-cK^{4/3}}$ upper obstruction.

Theorem 1 (Smooth binomial-bump obstruction). *Let $I, J \subset \mathbb{R}$ be finite open intervals such that $I \setminus \bar{J}$ contains a nonempty open interval. This includes every interval pair in Cases 3 and 4 of the source paper. There are constants $A, B, c > 0$ and real functions $f_m \in C_c^\infty(I)$ such that, for all sufficiently large integers m ,*

$$\frac{\|f'_m\|_{L^1(I)}}{\|f_m\|_{L^2(I)}} \leq Am^{3/4}, \quad (3)$$

and

$$\frac{\|Hf_m\|_{L^2(J)}}{\|f_m\|_{L^2(I)}} \leq Be^{-cm}. \quad (4)$$

Consequently, for suitable $B_0, c_0, K_0 > 0$,

$$\mathcal{S}_{I,J}(K) \leq B_0 e^{-c_0 K^{4/3}} \quad (K \geq K_0). \quad (5)$$

In particular, the conjectured inequality (1) is false, even on $C_c^\infty(I)$.

3 Proof

Choose a compact interval $K \Subset I \setminus \bar{J}$ with positive length and put

$$D = \text{dist}(K, J) > 0.$$

Choose $\rho > 0$ so small that $2\rho < |K|$ and $\rho < D$. Fix a point a such that $[a, a+2\rho] \subset K$, and fix a nonzero real $\phi \in C_c^\infty(0, 1)$. For $m \geq 2$, put $h_m = \rho/m$ and define

$$f_m(y) = \sum_{j=0}^m (-1)^j \binom{m}{j} \phi\left(\frac{y-a-jh_m}{h_m}\right). \quad (6)$$

All summands have disjoint supports, and their total span is $(m+1)h_m < 2\rho$. Thus f_m is real, nonzero, and belongs to $C_c^\infty(K) \subset C_c^\infty(I)$.

Exact norm identities

Disjointness and a change of variables give

$$\|f_m\|_{L^2(I)}^2 = h_m \|\phi\|_2^2 \sum_{j=0}^m \binom{m}{j}^2 = h_m \|\phi\|_2^2 \binom{2m}{m}, \quad (7)$$

while

$$\|f'_m\|_{L^1(I)} = \|\phi'\|_1 \sum_{j=0}^m \binom{m}{j} = 2^m \|\phi'\|_1. \quad (8)$$

Using the standard central-binomial estimate

$$\binom{2m}{m} \geq \frac{4^m}{2\sqrt{m}} \quad (m \geq 1), \quad (9)$$

we obtain

$$\begin{aligned} \frac{\|f'_m\|_1}{\|f_m\|_2} &\leq \sqrt{2} \frac{\|\phi'\|_1}{\|\phi\|_2} m^{1/4} h_m^{-1/2} \\ &= \sqrt{\frac{2}{\rho}} \frac{\|\phi'\|_1}{\|\phi\|_2} m^{3/4}. \end{aligned}$$

This proves (3).

Finite-difference cancellation

For $x \in J$, the support of f_m is separated from x , so no principal value is needed. Changing variables in every summand of (6) gives

$$\pi H f_m(x) = h_m \int_0^1 \phi(t) \sum_{j=0}^m (-1)^j \binom{m}{j} \frac{dt}{x - a - h_m(j+t)}. \quad (10)$$

For fixed x, t , let

$$q_{x,t}(s) = \frac{1}{x - a - h_m(s+t)}.$$

The alternating sum in (10) is $(-1)^m \Delta^m q_{x,t}(0)$. Repeated use of the fundamental theorem of calculus gives the exact identity

$$\Delta^m q(0) = \int_{[0,1]^m} q^{(m)}(u_1 + \cdots + u_m) du_1 \cdots du_m. \quad (11)$$

Every point $a + h_m(s+t)$ occurring here lies in K . Therefore

$$|q_{x,t}^{(m)}(s)| = \frac{m! h_m^m}{|x - a - h_m(s+t)|^{m+1}} \leq \frac{m! h_m^m}{D^{m+1}}. \quad (12)$$

Combining (10)–(12),

$$|H f_m(x)| \leq \frac{\|\phi\|_1}{\pi} \frac{m! h_m^{m+1}}{D^{m+1}} \quad (x \in J). \quad (13)$$

Taking the $L^2(J)$ norm and dividing by (7), then using (9), yields a constant $C = C(I, J, \phi)$ such that

$$\frac{\|Hf_m\|_{L^2(J)}}{\|f_m\|_{L^2(I)}} \leq C \frac{m! h_m^{m+1/2} m^{1/4}}{2^m D^{m+1}}. \quad (14)$$

Since $h_m = \rho/m$ and $m! \leq m^m$, the right-hand side is at most

$$C' m^{-1/4} \left(\frac{\rho}{2D} \right)^m. \quad (15)$$

Our choice $\rho < D$ makes the base smaller than $1/2$, proving (4).

Finally let A be the constant in (3). For sufficiently large K , choose an integer

$$m = \left\lfloor \left(\frac{K}{2A} \right)^{4/3} \right\rfloor.$$

Then $Am^{3/4} \leq K$ and $m \geq c'K^{4/3}$ after increasing the lower threshold. The definition (2) and (4) give (5).

If (1) held, then the same functions would satisfy

$$\frac{\|Hf_m\|_2}{\|f_m\|_2} \geq c_1 e^{-c_2 Am^{3/4}},$$

contradicting the geometric upper bound (15). This completes the proof. $\square$

4 Consequences and quantitative window

Corollary 2 (Negative answer to both source formulations). *For every Case 3 or Case 4 interval pair, the exponential conjecture after Theorem 2.1 and its weakly differentiable formulation (2.1) are false. The failure is not caused by jumps or roughness: it occurs inside $C_c^\infty(I)$.*

For $f \in W^{1,1}(I)$, total variation equals $\|f'\|_1$, so the theorem directly applies to the BV stability parameter used in Theorem 2.1. On the other hand, Theorem 2.4 of the source proves a lower bound of the form

$$\mathcal{S}_{I,J}(K) \geq c_1 e^{-c_2 K^2}.$$

Together with (5), the presently justified window is

$$c_1 e^{-c_2 K^2} \leq \mathcal{S}_{I,J}(K) \leq B_0 e^{-c_0 K^{4/3}} \quad (K \gg 1).$$

Determining the sharp exponent or non-power correction inside this window remains open. The present result is therefore a full refutation and a quantitative obstruction, not a full characterization of the optimal modulus.

5 Verification and novelty bounds

The accompanying script `code/verify_binomial_bumps.py` verifies the two binomial identities and the reciprocal-kernel finite-difference identity using exact integer/rational arithmetic. It also checks the $m^{3/4}$ scaling and geometric decay numerically. The computation is a sanity check and is not used in the proof.

On 9 August 2026, the four local run indexes were searched for the arXiv id, exact title, displayed conjecture, and core keyword combinations. Exact-title and keyword searches of arXiv and the web found the source, related truncated transform papers, and the dyadic model of Jaming–Pozzi–Wick [2], but no refutation, alternating-binomial construction, or $4/3$ obstruction. The dyadic work concerns a different operator and does not answer the present continuous conjecture. This is a bounded novelty check, not certification; MathSciNet, zbMATH, full citation graphs, and author correspondence were not exhaustively checked.

6 Human-review recommendation

Recommendation: promote after expert confirmation. The main review points are: (i) the source’s Case 4 orientation and the separated subinterval of $I \setminus \overline{J}$; (ii) the signs and derivative order in (11)–(12); (iii) the normalization in (14); and (iv) whether the construction has appeared outside the bounded novelty search. None of these checks revealed a defect in the present audit.

References

- [1] R. Alaifari, L. B. Pierce, and S. Steinerberger, *Lower bounds for the truncated Hilbert transform*, Rev. Mat. Iberoam. 32 (2016), no. 1, 23–56; arXiv:1311.6845; doi:10.4171/RMI/880.
- [2] P. Jaming, E. Pozzi, and B. D. Wick, *Lower bounds for the dyadic Hilbert transform*, Ann. Fac. Sci. Toulouse Math. (6) 26 (2017), no. 1, 185–204; arXiv:1605.05511.